Allen Tu

Ph.D. Student, Computer Science — University of Maryland
Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Biometrics, Generative Models

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Selected Publications

1. **Allen Tu***, H. Ying*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, 'SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping'. *CVPR*, 2026.
2. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, 'SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting'. *CVPR*, 2026.
3. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, 'TransFIRA: Transfer Learning for Face Image Recognizability Assessment'. *FG*, 2026.
4. A. Hanson*, **Allen Tu***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, 'PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting'. *CVPR*, 2025.
5. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, 'Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives'. *CVPR*, 2025.

* denotes equal contribution.

Education

University of Maryland, College Park College Park, MD
Ph.D. in Computer Science *January 2025 – May 2028 (Expected)*

- Advised by Professor Tom Goldstein; collaborating with Professor Matthias Zwicker

B.S./M.S. in Computer Science, Minor in Statistics *August 2019 – December 2024*

Research Experience

Amazon Bellevue, WA
Applied Science Intern *June 2026 – October 2026 (Expected)*

- Incoming Applied Science Intern on the Last Hundred Yards team, focusing on 3D reconstruction for robotics

University of Maryland Institute of Advanced Computer Studies College Park, MD
Graduate Research Assistant *August 2023 – Present*

- Developing scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA) program
- Created pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over $10\times$ fewer primitives while preserving visual fidelity [1, 4, 5]
- Introduced uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [2]

Systems & Technology Research Arlington, VA
Computer Vision Research Intern *May 2025 – January 2026*

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program
- Proposed an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [3]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at 1e-3 FMR using an encoder trained solely on public data [3]

Systems & Technology Research Arlington, VA
Computer Vision Research Intern *May 2024 – August 2024*

- Achieved 0.8566 correlation with ground-truth probe–to–gallery similarity and state-of-the art ERC performance for face image quality estimation (FIQA), outperforming prior methods on image-level evaluation [3]
- Designed a cluster-and-aggregate network that filters low-importance frame embeddings and fuses the remaining informative ones into a single video template, improving face recognition accuracy by up to 11.54%

Computer Vision Research Co-op *January 2023 – August 2023*

- Implemented a mixed-voting ensemble that improved open-set search performance by 24.24% for face, 39.18% for body, and 49.10% for fused multimodal recognition
- Trained a self-supervised Barlow Twins model robust to challenges like extreme distance variation and atmospheric interference, improving face recognition accuracy by 52.10% in severe operating conditions

Computer Vision Research Intern *June 2022 – August 2022*

- Integrated pose estimation, neural novel-view synthesis, semantic segmentation, and generative inpainting to synthesize realistic cross-subject garment transfers with 12.93 FID and strong biometric identity preservation
- Augmented real training data with garment-transfer images to overcome limited garment diversity and train clothing-invariant whole-body recognition models, improving robustness to cross-garment variation

Undergraduate Researcher College Park, MD
University of Maryland Department of Computer Science *January 2021 – December 2022*

- Analyzed 2.3 million dataset–visualization pairs to identify bias in machine learning visualization recommenders, linking model behavior to user-driven design patterns and proposing data-centric mitigation strategies [6]

Research Projects

IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA)
University of Maryland Institute of Advanced Computer Studies *August 2023 – Present*
Unconstrained 3D reconstruction and novel view synthesis in challenging real-world environments.

- Efficient rendering, compression, and training methods for 3D and 4D Gaussian Splatting (3DGS/4DGS) [1, 4, 5]
- Multi-view consistent super-resolution for training 3DGS with low-resolution imagery [2]
- Image, video, and multi-view diffusion model priors for sparse-view 3D reconstruction
- Uncertainty quantification algorithms for 3DGS and Neural Radiance Fields (NeRF) [4]

IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR)
Systems & Technology Research *June 2022 – January 2026*
Multimodal, opportunistic fusion of incomplete face, body, and gait information in severe operational conditions.

- Transfer Learning for Face Image Recognizability Assessment (2025) [3]
- Learned Frame Feature Aggregation for Face Recognition with Low-Quality Video (2024)
- Operating Condition-Invariant Barlow Twins and Multimodal Ensembling (2023)
- Style-Based Appearance Flow for Clothing-Robust Body Representation Learning (2022)

Service

SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models
Workshop Co-organizer *CVPR 2026*

- Co-organized the first CVPR workshop on security, privacy, and robustness in 3D generative vision models, coordinating peer review, securing $1K in industry sponsorship, and leading industry outreach

Professional Memberships: IEEE Member; Computer Vision Foundation (CVF); IEEE Biometrics Council

Teaching Experience

Peer Research Mentor College Park, MD
FIRE: Capital One Machine Learning *January 2021 – December 2022*

- Mentored 80 undergraduate students across two cohorts in a selective multi-semester machine learning research program, guiding students from foundational training to independent research project execution
- Designed and delivered a project-driven curriculum spanning data processing, model development, training, evaluation, and deployment using modern deep learning frameworks and recent research advances
- Advised student teams end-to-end on collaborative research projects, leading to presentations in areas including text-to-audio generation, image super-resolution, and face attribute recognition

Technical Skills

Languages: Python, CUDA, C/C++, C#, Java, SQL, Shell

Frameworks & Tools: PyTorch, TensorFlow, NumPy, SciPy, OpenCV, scikit-learn

Research Areas: 3D/4D Reconstruction, 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Differentiable Rendering, Biometrics, Diffusion Models, Vision Transformers (ViT/Swin), Uncertainty Quantification, Scalable Vision Systems, GPU Programming